

Information Visualization and Visual Analytics (M1522.000500)

Marks and Channels

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Why Marks and Channels?

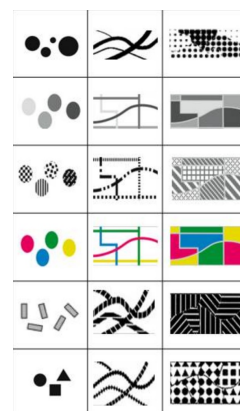
Why Marks and Channels?

- Every complex visual encoding **idiom** can be broken down into two orthogonal **components** : **Marks** and **Channels**

Idiom



Marks & Channels



Marks

- Basic graphical element
- Classified with their spatial dimension
 - Point (0D), Line (1D), Area (2D), Volume (3D)

➔ Points



➔ Lines



➔ Areas



Marks and Their Types

- Geometric primitives

Marks as Items/Nodes (Marks to represent individual items)

➔ Points



➔ Lines



➔ Areas

**Marks as Links** (Marks to represent links between items)

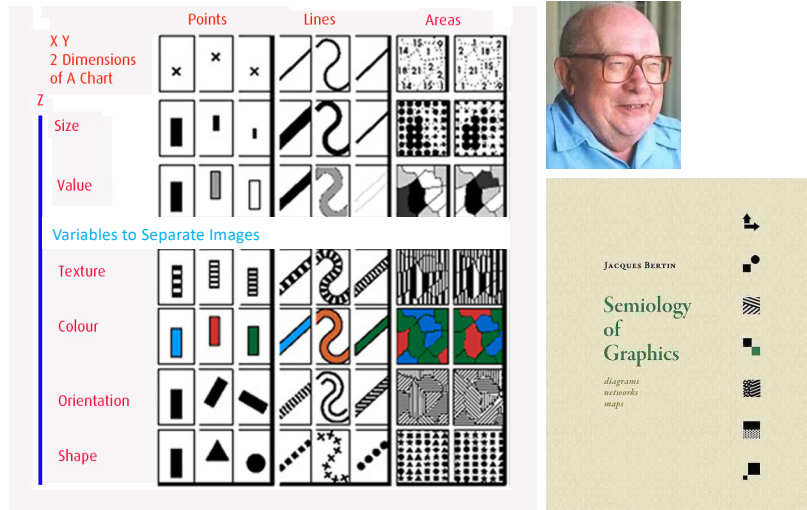
➔ Containment



➔ Connection



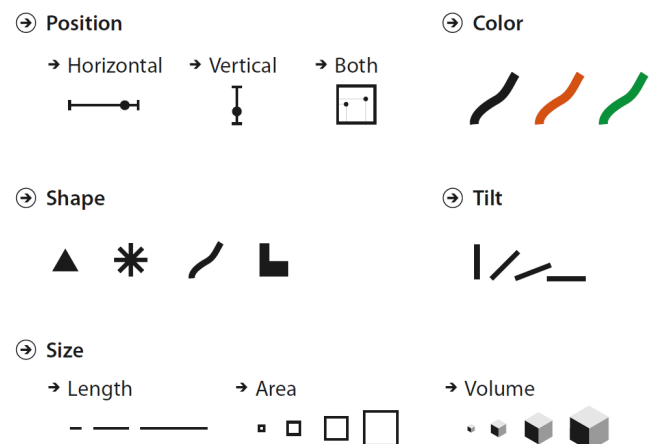
Variables of a Visual Image



- A way to control the **appearance** of marks

- human perceptual system has two fundamentally different kinds of sensory modalities

- **Identity** channel (what, where)
- **Magnitude** channel (how much)



Marks and Channels

- **Marks**

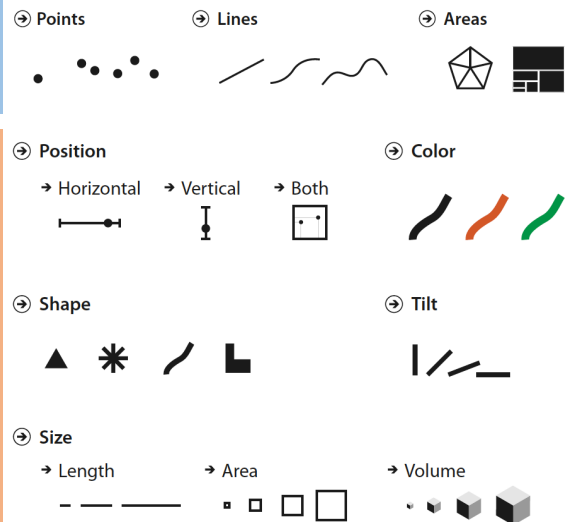
- geometric primitives

- **Channels**

- control appearance of marks
- can redundantly code with multiple channels

- Interactions (marks vs. size and shape channels)

- size and shape channels cannot be used on all types of marks
 - constraints that arise from the way that marks are defined
- **area** marks *fully constrained* (e.g., an area mark denoting a state or province)
 - **cannot be size or shape coded**
- **line** marks convey position and length
 - **can only be size coded in 1D** (width)
- **point** marks only convey position; **no area constraints**
 - **can be size and shape coded**



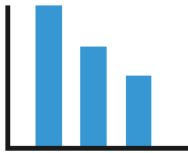
Visual Encoding

Analyze Idiom Structure



Analyze Idiom Structure

- as combination of marks and channels



1:
vertical position

mark: line



2:
vertical position
horizontal position

mark: point



3:
vertical position
horizontal position
color hue

mark: point



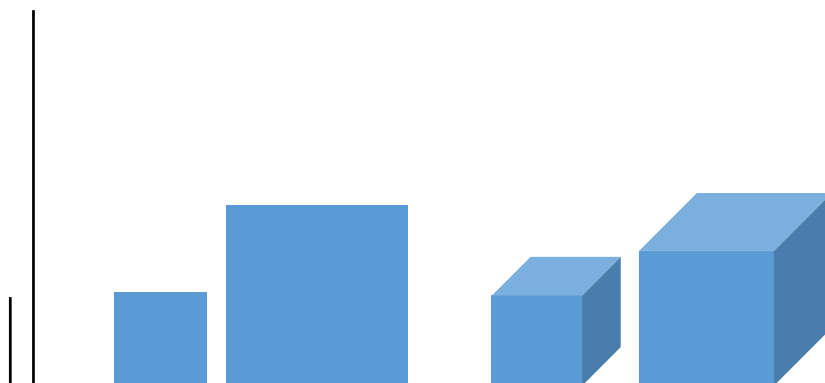
4:
vertical position
horizontal position
color hue
size (area)

mark: point

1D, 2D, and 3D

1D, 2D, and 3D

- size ratio for each pair → 1:4



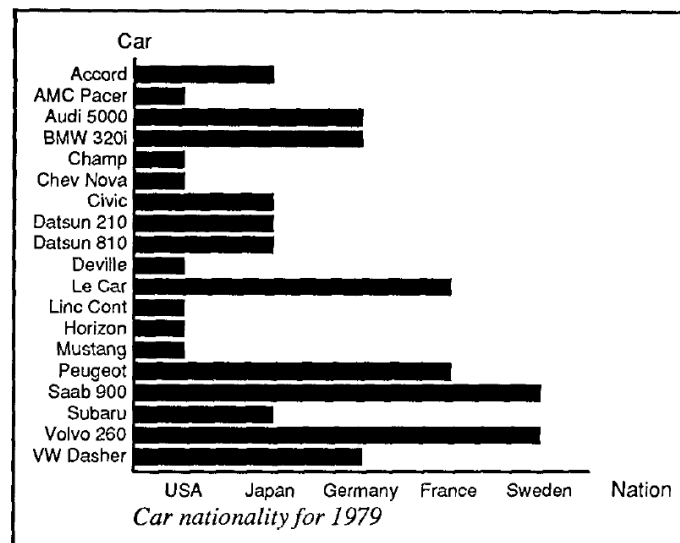
Expressiveness and Effectiveness

- Expressiveness
 - Vis idiom should express **all** of, and **only**, the information in the dataset attributes
- Effectiveness
 - Most important attributes should be encoded with the most effective channels

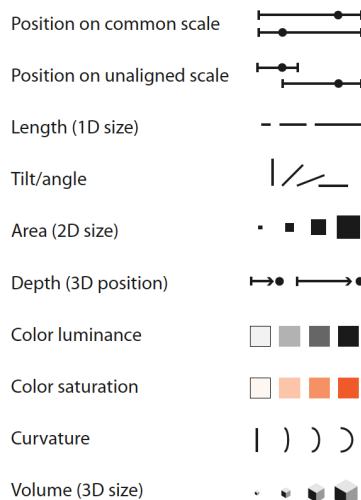
Expressiveness and Effectiveness

- expressiveness principle
 - match channel and data characteristics
- effectiveness principle
 - encode most important attributes with highest ranked channels
- rankings: where do they come from?
 - accuracy
 - discriminability
 - separability
 - popout

Any Better Encoding?



Channels: Expressiveness types and effectiveness rankings

➡ **Magnitude Channels: Ordered Attributes**➡ **Identity Channels: Categorical Attributes**• **Magnitude Channels for Ordered Attributes**

- Position (aligned scale > unaligned) > Length (1D size) > Tilt/angle > Area (2D size) > Depth > Color (luminance = saturation) > Curvature = Volume (3D size)

• **Identity Channels for Categorical Attributes**

- Spatial region > Color Hue > Motion > Shape

• **Position dominates the user's mental model**

Accuracy: Fundamental Theory

Stevens' Power Law

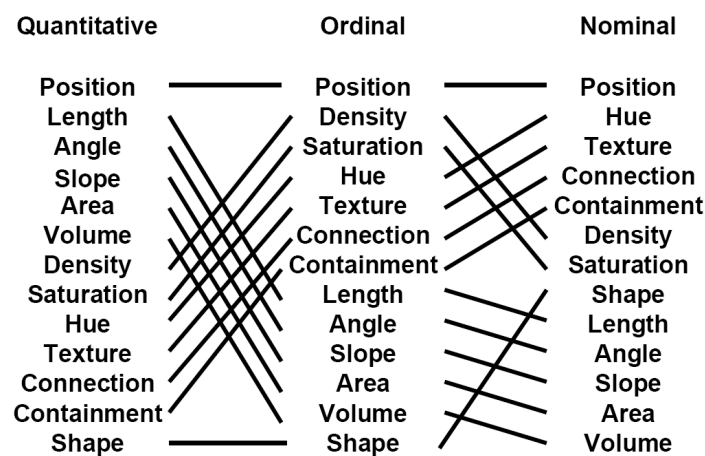
- p : perceived magnitude
- a : actual magnitude
- $p = ka^{\alpha}$
- $p_1/p_2 = (a_1/a_2)^{\alpha}$
- length judgment: $\alpha \approx 1$
- area judgment: $\alpha < 1$
- volume judgment: $\alpha \ll 1$

Continuum	Exponent (α)	Stimulus condition
Loudness	0.67	Sound pressure of 3000 Hz tone
Vibration	0.95	Amplitude of 60 Hz on finger
Vibration	0.6	Amplitude of 250 Hz on finger
Brightness	0.33	5° target in dark
Brightness	0.5	Point source
Brightness	0.5	Brief flash
Brightness	1	Point source briefly flashed
Lightness	1.2	Reflectance of gray papers
Visual length	1	Projected line
Visual area	0.7	Projected square
Redness (saturation)	1.7	Red-gray mixture
Taste	1.3	Sucrose
Taste	1.4	Salt
Taste	0.8	Saccharin
Smell	0.6	Heptane
Cold	1	Metal contact on arm
Warmth	1.6	Metal contact on arm
Warmth	1.3	Irradiation of skin, small area
Warmth	0.7	Irradiation of skin, large area
Discomfort, cold	1.7	Whole body irradiation
Discomfort, warm	0.7	Whole body irradiation
Thermal pain	1	Radiant heat on skin
Tactile roughness	1.5	Rubbing emery cloths
Tactile hardness	0.8	Squeezing rubber
Finger span	1.3	Thickness of blocks
Pressure on palm	1.1	Static force on skin
Muscle force	1.7	Static contractions
Heaviness	1.45	Lifted weights
Viscosity	0.42	Stirring silicone fluids
Electric shock	3.5	Current through fingers
Vocal effort	1.1	Vocal sound pressure
Angular acceleration	1.4	5 s rotation
Duration	1.1	White noise stimuli

Accuracy: Fundamental Theory

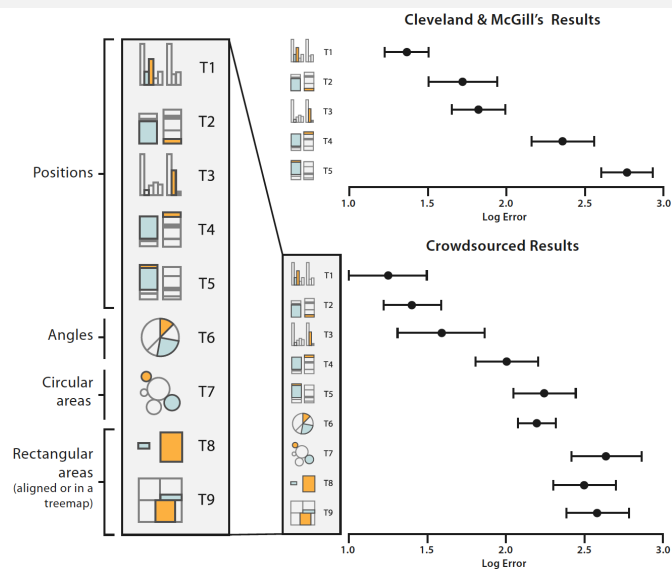
Visual Encoding Principles according to accuracy

- Channel Ranking Varies by Data Type



Crowdsourced Results

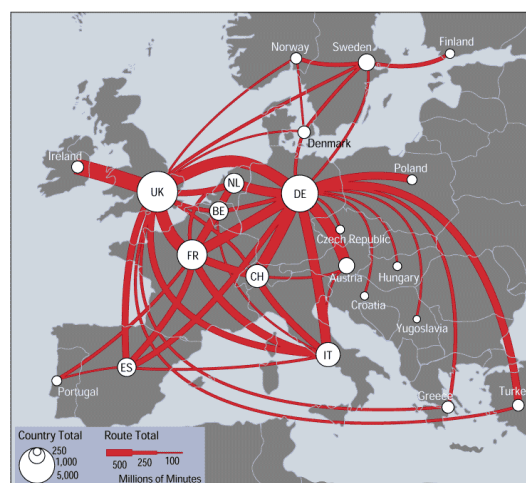
- T1~T3
 - position along a common scale
- T4~T5
 - length encoding
- T6: angle
- T7: circular area
- T8~T9
 - rectangular area



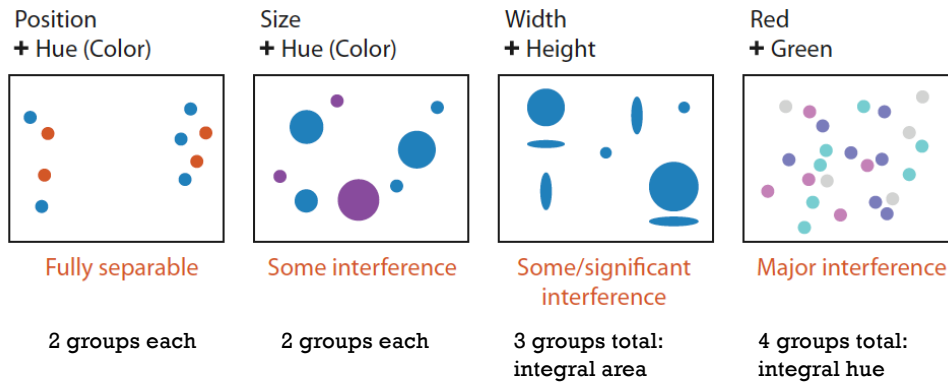
Discriminability of a Channel

How many usable steps?

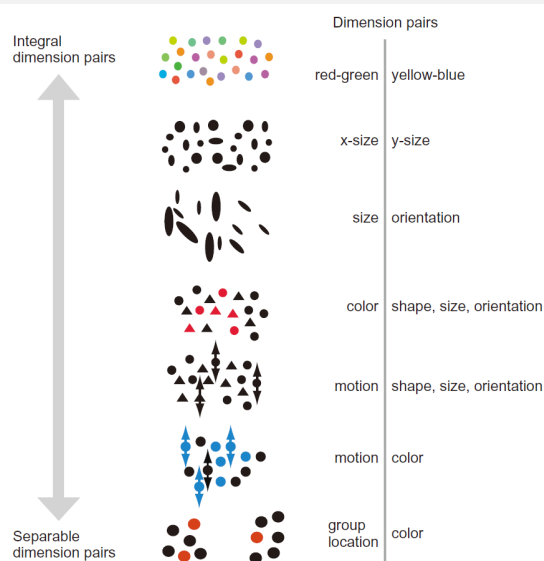
- linewidth: only a few



Separable channels vs. Integral channels



Separability

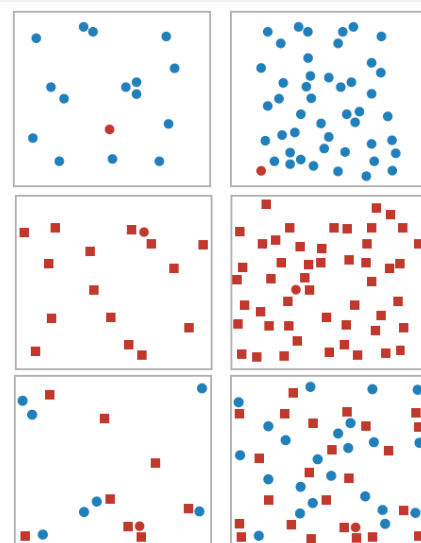


Preattentive Processing

- Cognitive operations done preattentively, without the need for focused attention
 - less than 200-250 ms
 - eye movements take 200 ms
 - minimum time to initiate eye movement
 - involves only information available in a single glance
- Popout effects
- Segmentation effects

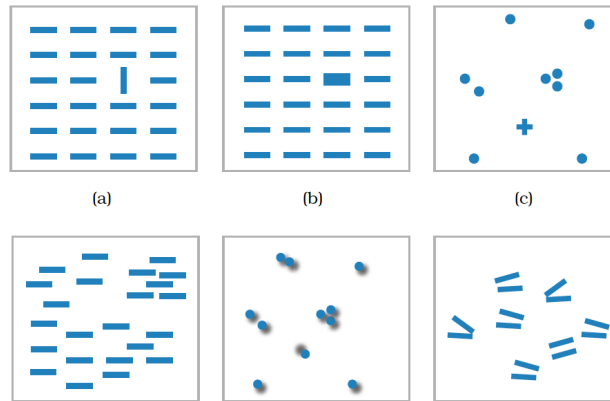
Preattentive Perception

- find the red dot
 - how long does it take?
- **parallel** processing on many individual channels
 - speed *independent* of # of distractors
 - speed *depends* on channel and amount of difference from distractors
- serial search for (almost all) combinations
 - speed depends on number of distractors

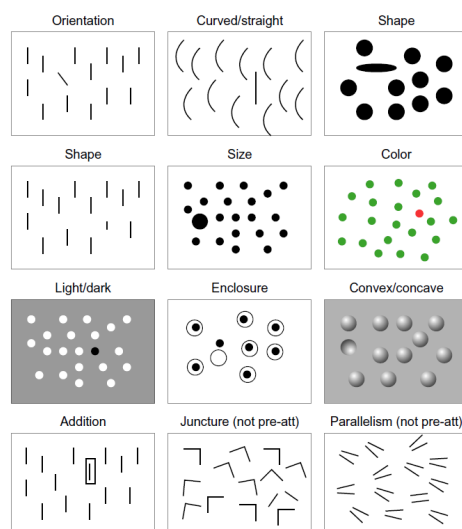


Popout

- many channels: tilt, size, shape, proximity, shadow direction, ...
- but not all! parallel line pairs do not pop out from tilted pairs



Popout



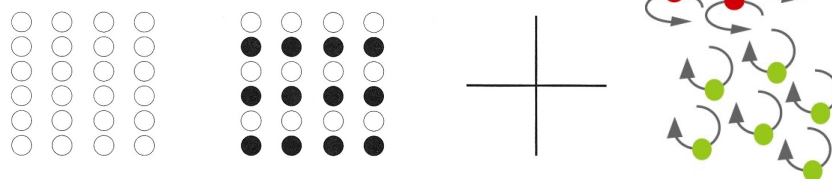
Gestalt Psychology

- Principles of perceptual organization
 - the **whole** is different from the **sum of its parts**
 - how smaller objects are grouped to form larger ones
 - "gestalt": German for "pattern/form/shape"
 - "leaving us with a set of *descriptive* principles, but without a model of perceptual processing"
 - rules themselves still very useful
- Law of Prägnanz
 - law of simplicity, law of good figure
 - fundamental principle of gestalt perception
 - tend to order our experience in a manner that is regular, orderly, symmetric, and simple
 - simplest possibility wins

Gestalt Principles

How do we perceive groups

- **Proximity**: tendency of elements to be associated with nearby elements
- **Similarity**: tendency of elements to be associated with similar elements
- **Continuity**: preference for continuous, unbroken, smoothest contours with the *simplest possible physical explanation*
- **Common Fate**: things moving together



Gestalt Psychology

- containment
- connection
- proximity
 - same spatial region
- similarity
 - same values as other categorical channels

Marks as Links

➔ Containment



➔ Connection



➔ Identity Channels: Categorical Attributes

Spatial region



Color hue



Motion



Shape

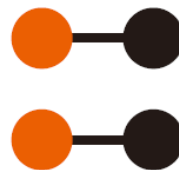


Channel Effectiveness

Gestalt laws - Connectedness



(a)



(b)

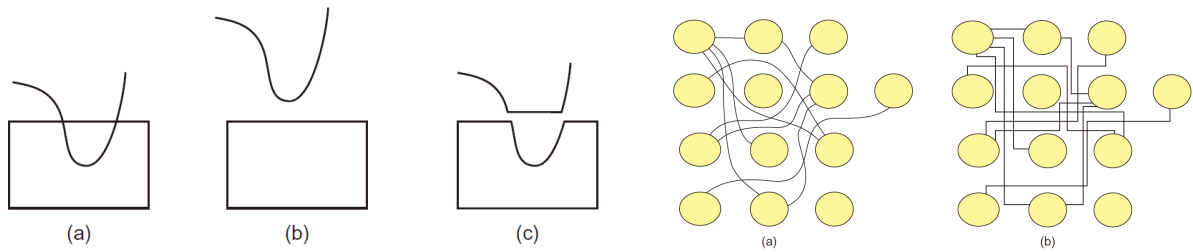


(c)



(d)

Gestalt laws - Continuity



Principles of Graphic Design

- CRAP: Contrast, Repetition, Alignment, and Proximity
- Alignment, Repetition, Contrast, Proximity, Balance
- Balance, Alignment, Hierarchy, Contrast, Rhythm, Proximity, Color and Space
- Unity, Emphasis, Scale, Balance

Contrast

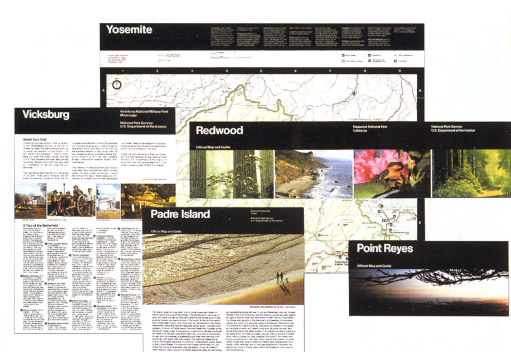
- guides the user's attention to the key elements of a design
- maintain the distinction between similar elements in a design
 - Dark vs Light
 - Thick vs Thin
 - Contemporary vs Traditional
 - Large vs Small
- related to "Preattentive Processing"



<https://www.lcca.org.uk/blog/careers/7-basic-principles-of-graphic-design/>

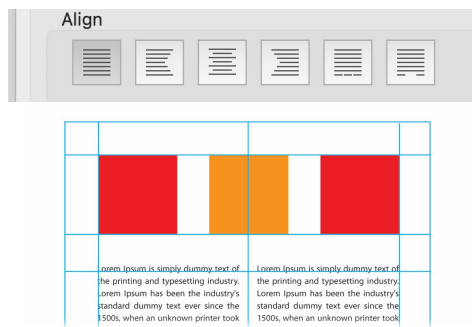
Repetition

- elements being copied or mimicked numerous times
 - improve unity and harmony
- creates visual **consistency** in page designs
 - using the same style of headlines,
 - the same style of initial capitals, or
 - repeating the same basic layout from one page to another



Alignment

- the act of keeping design objects in line
 - not only vertically or horizontally, but across any linear plane
- creates a visual connection between elements such as images, shapes or blocks of texts



<https://designartlovestory.wordpress.com/2011/09/08/the-principles-of-design-alignment/>

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Information Visualization and Visual Analytics – Marks & Channels

Alignment by Spatial Position

➡ **Magnitude Channels: Ordered Attributes**

Position on common scale	
Position on unaligned scale	
Length (1D size)	
Tilt/angle	
Area (2D size)	
Depth (3D position)	
Color luminance	
Color saturation	
Curvature	
Volume (3D size)	

➡ **Identity Channels: Categorical Attributes**

Spatial region	
Color hue	
Motion	
Shape	

- **Magnitude** = Prothetic = **How much**
- **Identity** = Metathetic = **What-Where**

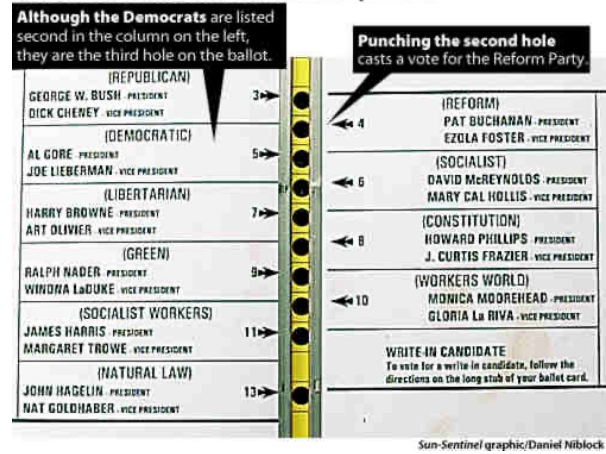
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Information Visualization and Visual Analytics – Marks & Channels

False Alignment

- Butterfly Ballot
- Florida 2000

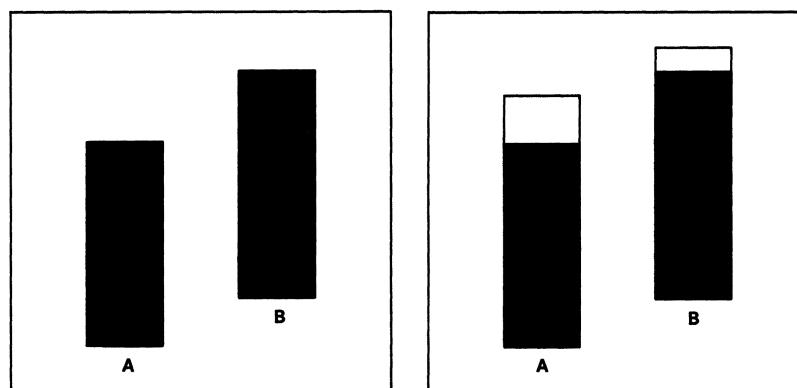
Confusion over Palm Beach County ballot



<http://www.asktog.com/columns/042ButterflyBallot.html>

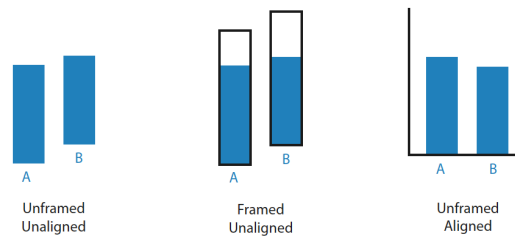
Relative Perception

Which is Longer?



Weber's Law

- Perceptual system mostly operates with **relative** judgements, **not absolute**
 - that's why accuracy increases with common frame/scale and alignment
- Weber's Law: the **perceived change** in stimuli is proportional to the **initial stimuli**
 - ratio of increment to background is constant $\frac{(JND)dS}{S} = \text{constant}$
 - filled rectangles are long and differ in length by 15% → difficult judgement
 - white rectangles are short and differ in length by 50% → easy judgement



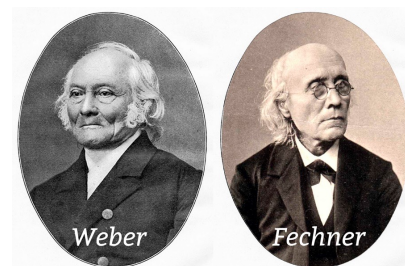
after [Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods, Cleveland and McGill, Journ. American Statistical Association 79:387 (1984), 531-554.]

Information Visualization and Visual Analytics – Marks & Channels

Weber-Fechner's Law

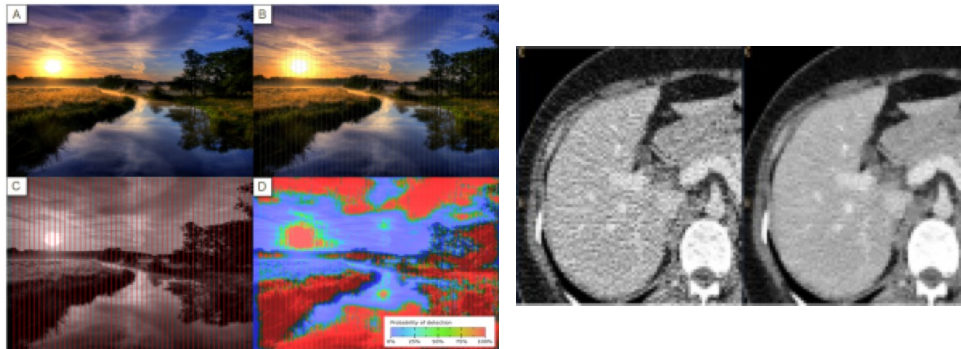
- $\frac{\delta I}{I} = K$, where I is a stimulus intensity and K is a fixed percentage
- **perceived change** in stimuli is proportional to **initial stimuli**
- **detectable difference** in stimulus intensity I as a fixed percentage K of the object magnitude

- **JND**: just noticeable difference



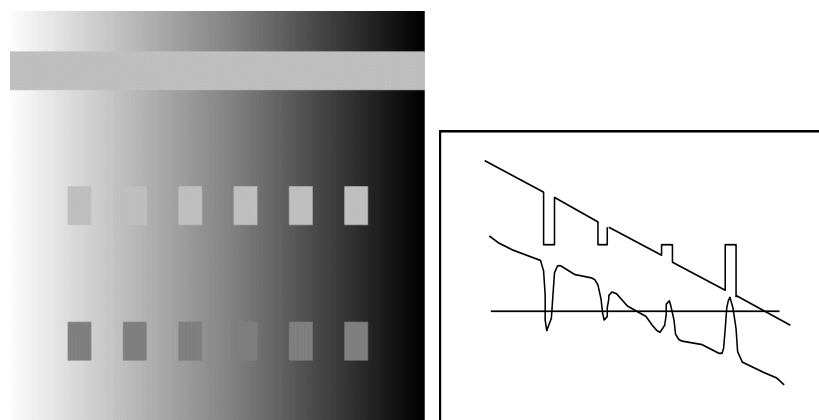
JND and Visual Perception

- Contrast sensitivity function
 - Sensitivity of visual perception depends on frequency ((related to saliency))



Luminance contrast – Simultaneous Brightness Contrast

- Luminance perception is based on relative judgements



Contrast for constancy

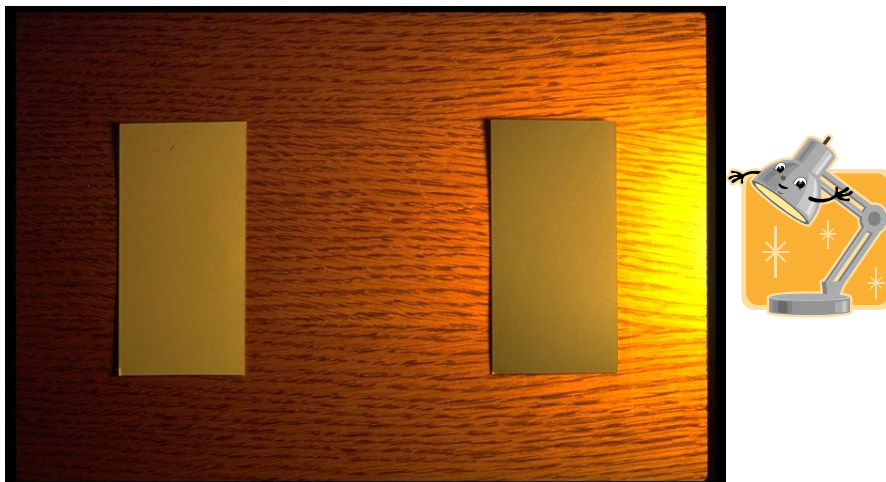
- how the visual system discounts illumination to perceive consistent surface colors

- Similar Visual Perception?



Contrast for constancy

- Does mild gray paper reflect about the same amount of light as the white paper?



Illumination Discounted

- In bright sunlight or moonlight, we can tell whether a surface is black, white, or gray
- Luminance is **completely unrelated** to perceived lightness (or brightness)
 - black paper in full sunlight – 1000 candelas
 - white paper in an office – 50 candelas
- Visual system extracts surface information, not raw luminance
- Discounts illumination level
 - Intensity (illumination level)
 - Spectrum (illumination color)
- Mechanisms
 - Photoreceptor **adaptation**: rods and cones adjust sensitivity (like a camera auto-exposure).
 - **Relative coding**: the brain interprets brightness relative to surrounding surfaces (e.g., ratio-taking).
 - **Perceptual constancy**: we perceive the white wall as “white” even though its luminance drops drastically at dusk.

Note

- Questions?